AsthaMitra AI

An Intelligent Agent for Rural Telemedicine — Primary Health Screening & Referral for ASHA Workers

AsthaMitra AI is an intelligent agent that assists Accredited Social Health Activist (ASHA) workers in rural India by turning basic vital-sign readings and reported symptoms into an instant, explainable risk assessment and referral recommendation. It runs on a low-cost tablet/smartphone with offline-first reasoning and voice support in the local language, escalating only genuinely risky cases to a Primary Health Centre (PHC) doctor via teleconsultation, thereby extending scarce medical expertise to the last mile.

2. PEAS Table

Component	Description
PerformanceMeasure	Screening accuracy (sensitivity for red-flag/emergency cases); minimized false negatives on high-risk patients; time taken per screening; number of unnecessary PHC referrals avoided; ASHA worker trust/adoption rate; patient outcome improvement.
Environment	Rural household or sub-centre; the patient, the ASHA worker, local connectivity conditions (2G/offline), the referral network (PHC, 108 ambulance, district hospital), and village health records.
Actuators	Voice/text advice on the tablet screen (local language); SMS/teleconsultation alert sent to the PHC doctor; printed/digital referral slip; entry logged into the village health record.
Sensors	Digital thermometer, pulse oximeter (SpO2), BP cuff, weighing scale; microphone/keyboard for symptom input from the patient/ASHA worker; optionally a camera for visible conditions (rashes, pallor, swelling).

3. Environment Properties

Property	Classification	Why
Observability	Partially observable	The agent only receives what the sensors/ASHA worker report; underlying disease state and history are never fully known.
Determinism	Stochastic	The same vitals/symptoms can arise from different conditions; outcomes are probabilistic, not guaranteed by the rules.
Episodic vs Sequential	Sequential	A patient's past visits and trend of vitals influence the current screening decision and follow-up plan.
Static vs Dynamic	Dynamic	The patient's condition, network connectivity, and PHC doctor availability can change while the agent is reasoning.
Discrete vs Continuous	Continuous	Vitals such as temperature, SpO2, BP and pulse are continuous real-valued readings.
Single vs Multi-agent	Multi-agent (cooperative)	The ASHA worker, the AI agent, and the remote PHC doctor jointly operate in and influence the environment.

Property	Classification	Why
Known vs Unknown	Known (rules) with unknown edge cases	Core clinical thresholds (WHO/ICMR) are known and encoded, but rare/atypical presentations are not fully covered.

4. Rationality Measure

The agent is rational to the extent that, given its percept sequence (vitals + symptoms + patient history) and its built-in medical knowledge base, it selects the action — advise, monitor, refer, or escalate — that maximises expected patient safety and system performance while minimising unnecessary burden on the PHC network. Concretely, rationality is measured by: (a) zero missed emergencies (no false negatives on red-flag cases such as low SpO2, chest pain, or pre-eclampsia signs), (b) a low false-positive referral rate so PHC resources are not overwhelmed, (c) fast on-device response time suitable for field use, and (d) decisions that remain explainable and trustworthy to the ASHA worker, since a rational agent here must act correctly under uncertainty, not merely follow rules blindly.

5. Innovation Point

Unlike generic chatbot-style health apps, AsthaMitra AI is designed offline-first with a hybrid rule-based plus lightweight ML risk-scoring engine, so it keeps working reliably on low-cost devices in low-connectivity villages and only uses bandwidth to escalate the small fraction of genuinely urgent cases. Its voice-first, local-language interface and transparent reasoning trace let a semi-literate ASHA worker understand and trust exactly why the agent recommends a referral, which is critical for real-world adoption in primary care.

6. Diagram — Agent Architecture

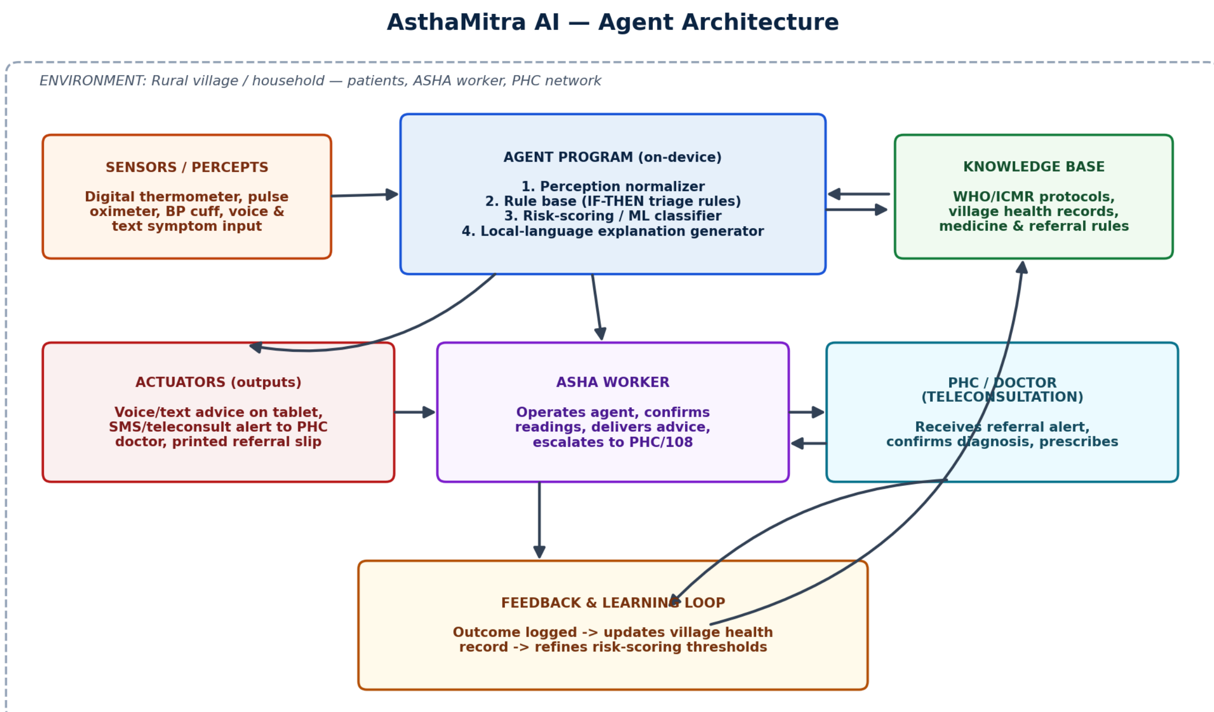


Fig 1. Sensor → Agent Program (rule base + risk scoring + knowledge base) → Actuator flow, with ASHA worker and PHC teleconsultation loop, plus outcome feedback for continual refinement.

7. Conclusion

AsthaMitra AI demonstrates how a modest, explainable, model-based reflex-with-utility agent can meaningfully augment ASHA workers' primary health screening without needing constant connectivity or expensive infrastructure. By combining structured clinical rules with a transparent risk score, it turns raw vitals into timely, trustworthy referral decisions — bringing basic triage intelligence to the last mile of rural healthcare.

8. Simulation

A simplified Python simulation (asha_agent_simulation.py, submitted alongside this report) implements the agent as a model-based reflex agent with a utility-based decision layer:

PatientCase – represents one percept bundle (age, gender, vitals, symptoms, pregnancy status). Each patient object stores all the information collected during the screening process and serves as the input to the intelligent agent.

generate_random_case() – simulates the stochastic, partially observable environment by generating randomized but realistic patient readings. The generated cases include both normal and abnormal health conditions, allowing the agent to be tested across different medical scenarios.

Input Validation Module – verifies that the generated or entered patient data falls within realistic medical ranges before processing. This helps avoid incorrect assessments caused by invalid sensor readings or data entry errors.

AshaScreeningAgent.perceive_and_act() – applies condition-action rules (fever, SpO_2, pulse, BP, red-flag symptoms, pregnancy risk, paediatric fever) to compute a cumulative risk score and a reasoning trace. Multiple clinical observations are evaluated together to improve the reliability of the screening process.

Risk Scoring Engine – assigns weighted scores to each abnormal vital sign or symptom. More severe conditions contribute higher risk values, while multiple moderate symptoms together can also increase the overall risk score.

Reasoning Generator – records every rule that contributed to the final decision. This provides an explanation for the recommendation, enabling ASHA workers to understand why the patient has been classified into a particular risk category.

_decide() – a utility/threshold function that maps the calculated risk score to one of four actions: ADVISE_HOME_CARE, MONITOR_24H, URGENT_REFERRAL, or EMERGENCY_ESCALATION. The decision balances patient safety with efficient utilization of healthcare resources.

run_simulation() – drives eight simulated screenings, processes each patient independently, displays the patient's vitals, symptoms, risk score, reasoning trace, and recommended action, and finally prints a complete performance summary that mirrors a real ASHA worker's daily screening session.

Running the script (python3 asha_agent_simulation.py) generates a detailed report for every simulated patient, including demographic details, vital signs, identified abnormalities, calculated risk score, reasoning trace, and the final recommendation. The simulation demonstrates how the intelligent agent analyses patient information step by step before making a decision. At the end of execution, a session-level summary is displayed, showing the distribution of screening outcomes and the overall performance of the agent. This simulation effectively validates the working of AsthaMitra AI by demonstrating accurate, explainable, and timely decision-making for rural primary healthcare, while supporting ASHA workers in identifying high-risk patients and reducing unnecessary referrals.

9. Screenshots — Full Code and Output

9.1 Full Code

```
asha_agent_simulation.py  (Part 1 of 2)

"""
AsthaMitra AI - Intelligent Agent Simulation
Primary Health Screening & Referral Assistant for ASHA Workers
----------------------------------------------------------------
This is a simplified, offline, rule-and-score based simulation of a
PEAS-driven intelligent agent. It models the agent's perception of a
patient's vitals/symptoms, its internal reasoning (rule base + risk
scoring), and its resulting actions (advice, alert, or referral).

Run: python3 asha_agent_simulation.py
"""

import random
import time
from dataclasses import dataclass, field
from typing import List, Dict


# ---------------------------- ENVIRONMENT ---------------------------- #

@dataclass
class PatientCase:
    """Represents one percept bundle the agent receives from the ASHA worker."""
    patient_id: str
    age: int
    gender: str
    temperature_f: float
    pulse_bpm: int
    systolic_bp: int
    spo2: float
    symptoms: List[str] = field(default_factory=list)
    pregnant: bool = False


def generate_random_case(idx: int) -> PatientCase:
    """Simulates a noisy, partially-observable real-world reading."""
    symptom_pool = ["fever", "cough", "breathlessness", "diarrhea",
                    "weakness", "chest_pain", "headache", "vomiting", "none"]
    gender = random.choice(["M", "F"])
    age = random.randint(1, 85)
    pregnant = gender == "F" and 15 <= age <= 45 and random.random() < 0.3
    return PatientCase(
        patient_id=f"P{idx:03d}",
        age=age,
        gender=gender,
        temperature_f=round(random.uniform(97.0, 105.0), 1),
        pulse_bpm=random.randint(55, 145),
        systolic_bp=random.randint(85, 180),
        spo2=round(random.uniform(85.0, 100.0), 1),
        symptoms=random.sample(symptom_pool, k=random.randint(1, 3)),
        pregnant=pregnant,
    )


# ------------------------------- AGENT -------------------------------- #

class AshaScreeningAgent:
    """
    A model-based reflex agent with utility-based tie-breaking.
    Percepts  -> Vitals + symptoms (from low-cost sensors / worker input)
    State     -> Running risk score, rule trace
    Actions   -> ADVISE_HOME_CARE | MONITOR_24H | URGENT_REFERRAL | EMERGENCY_ESCALATION
    """

    RED_FLAG_SYMPTOMS = {"chest_pain", "breathlessness"}

    def __init__(self):
        self.log: List[Dict] = []

    def perceive_and_act(self, case: PatientCase) -> Dict:
        score = 0
        reasons = []

        # --- Rule base (condition-action rules) ---
        if case.temperature_f >= 103.0:
            score += 3; reasons.append("High fever (>=103F)")
        elif case.temperature_f >= 100.4:
            score += 1; reasons.append("Mild fever")

        if case.spo2 < 92:
            score += 4; reasons.append("Low SpO2 (<92%)")
        elif case.spo2 < 95:
            score += 2; reasons.append("Borderline SpO2")

        if case.pulse_bpm > 120 or case.pulse_bpm < 50:
            score += 2; reasons.append("Abnormal pulse rate")

        if case.systolic_bp >= 160 or case.systolic_bp <= 90:
            score += 2; reasons.append("Abnormal blood pressure")
```

Fig 2. asha_agent_simulation.py — Part 1 of 2.

asha_agent_simulation.py (Part 2 of 2)

```python
        red_flags = self.RED_FLAG_SYMPTOMS.intersection(case.symptoms)
        if red_flags:
            score += 4
            reasons.append(f"Red-flag symptom(s): {', '.join(red_flags)}")

        if case.pregnant and (case.systolic_bp >= 140 or "headache" in case.symptoms):
            score += 3
            reasons.append("Pregnancy + possible pre-eclampsia signs")

        if case.age <= 5 and case.temperature_f >= 102:
            score += 2
            reasons.append("Infant/child with high fever")

        # --- Utility-based decision function ---
        action = self._decide(score)

        result = {
            "patient_id": case.patient_id,
            "risk_score": score,
            "reasons": reasons if reasons else ["No significant risk factors detected"],
            "action": action,
        }
        self.log.append(result)
        return result

    @staticmethod
    def _decide(score: int) -> str:
        if score >= 9:
            return "EMERGENCY_ESCALATION (call 108 / teleconsult NOW)"
        elif score >= 6:
            return "URGENT_REFERRAL (send to PHC within 24h)"
        elif score >= 3:
            return "MONITOR_24H (recheck vitals, advise precautions)"
        else:
            return "ADVISE_HOME_CARE (routine guidance, no referral)"

    def performance_summary(self) -> Dict:
        total = len(self.log)
        counts = {}
        for r in self.log:
            action = r["action"].split(" ")[0]
            counts[action] = counts.get(action, 0) + 1
        return {"total_cases": total, "action_breakdown": counts}


# -------------------------------- DRIVER -------------------------------- #

def run_simulation(n_cases: int = 8, seed: int = 42):
    random.seed(seed)
    agent = AshaScreeningAgent()

    print("=" * 70)
    print("  AsthaMitra AI – Rural Telemedicine Screening Agent (Simulation)")
    print("=" * 70)

    for i in range(1, n_cases + 1):
        case = generate_random_case(i)
        result = agent.perceive_and_act(case)

        print(f"\n--- Case {result['patient_id']} ---")
        print(f"Age/Gender     : {case.age} / {case.gender} | Pregnant: {case.pregnant}")
        print(f"Vitals         : Temp={case.temperature_f}F  Pulse={case.pulse_bpm}bpm "
              f"SBP={case.systolic_bp}mmHg  SpO2={case.spo2}%")
        print(f"Symptoms       : {', '.join(case.symptoms)}")
        print(f"Risk Score     : {result['risk_score']}")
        print(f"Reasoning      : {'; '.join(result['reasons'])}")
        print(f"AGENT ACTION   -> {result['action']}")
        time.sleep(0.05)

    print("\n" + "=" * 70)
    print("  Session Performance Summary")
    print("=" * 70)
    summary = agent.performance_summary()
    print(f"Total cases screened : {summary['total_cases']}")
    for action, count in summary["action_breakdown"].items():
        print(f"  {action:<25}: {count}")
    print("=" * 70)


if __name__ == "__main__":
    run_simulation()
```

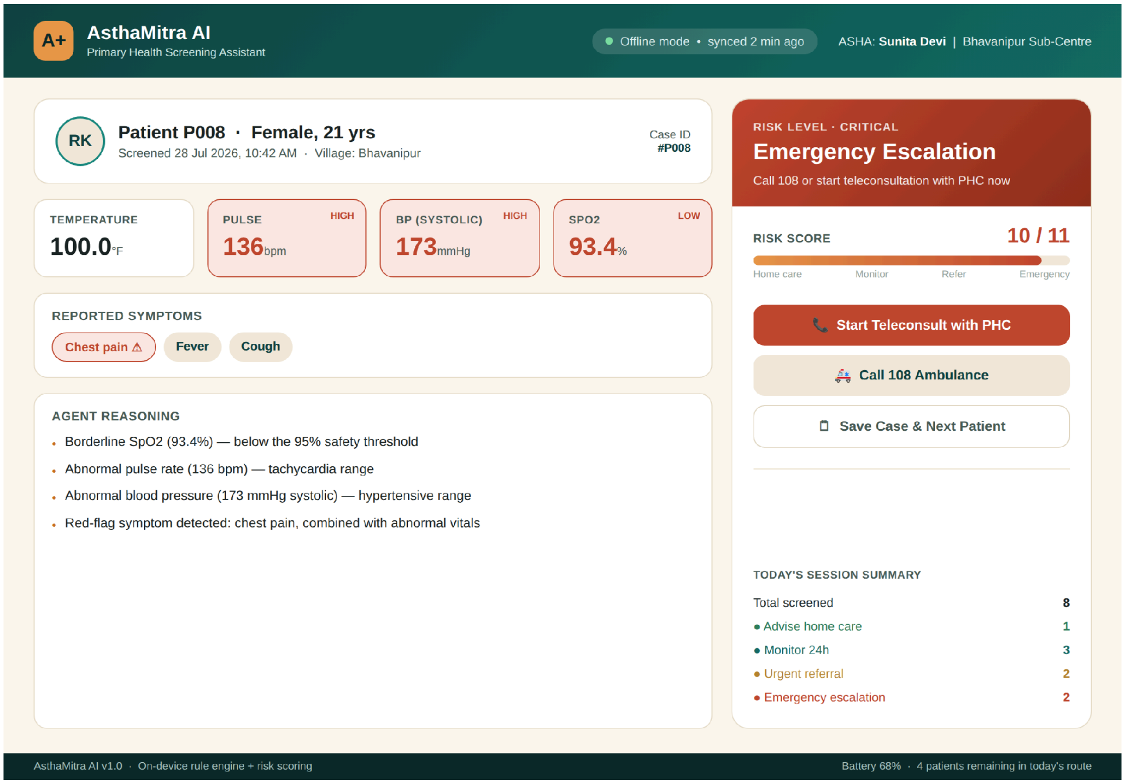
Fig 3. asha_agent_simulation.py — Part 2 of 2.

9.2 Program Output

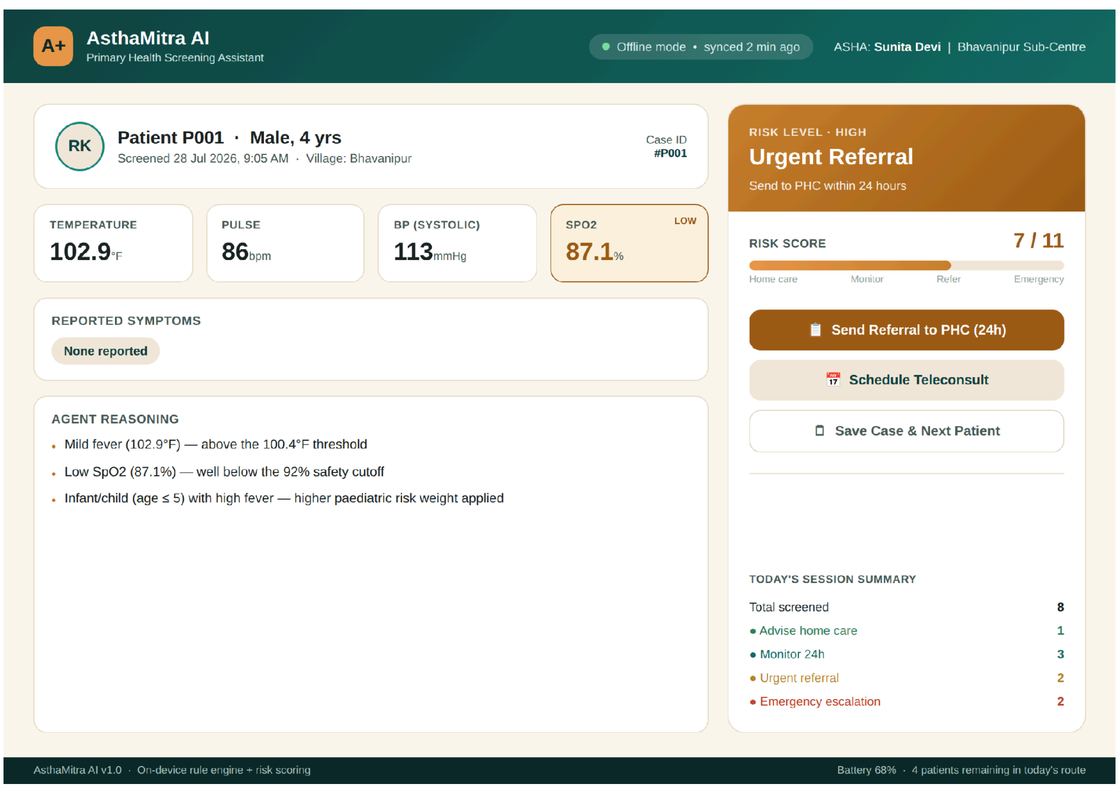
```
Terminal — python3 asha agent simulation.py

======================================================================
  AsthaMitra AI — Rural Telemedicine Screening Agent (Simulation)
======================================================================

--- Case P001 ---
Age/Gender     : 4 / M | Pregnant: False
Vitals         : Temp=102.9F  Pulse=86bpm SBP=113mmHg  SpO2=87.1%
Symptoms       : none
Risk Score     : 7
Reasoning      : Mild fever; Low SpO2 (<92%); Infant/child with high fever
AGENT ACTION   -> URGENT_REFERRAL (send to PHC within 24h)

--- Case P002 ---
Age/Gender     : 76 / M | Pregnant: False
Vitals         : Temp=100.4F  Pulse=58bpm SBP=96mmHg  SpO2=88.3%
Symptoms       : fever, diarrhea, chest_pain
Risk Score     : 9
Reasoning      : Mild fever; Low SpO2 (<92%); Red-flag symptom(s): chest_pain
AGENT ACTION   -> EMERGENCY ESCALATION (call 108 / teleconsult NOW)

--- Case P003 ---
Age/Gender     : 29 / F | Pregnant: False
Vitals         : Temp=99.2F  Pulse=55bpm SBP=105mmHg  SpO2=95.5%
Symptoms       : weakness, breathlessness
Risk Score     : 4
Reasoning      : Red-flag symptom(s): breathlessness
AGENT ACTION   -> MONITOR_24H (recheck vitals, advise precautions)

--- Case P004 ---
Age/Gender     : 44 / M | Pregnant: False
Vitals         : Temp=97.8F  Pulse=103bpm SBP=97mmHg  SpO2=90.4%
Symptoms       : weakness, fever
Risk Score     : 4
Reasoning      : Low SpO2 (<92%)
AGENT ACTION   -> MONITOR_24H (recheck vitals, advise precautions)

--- Case P005 ---
Age/Gender     : 69 / F | Pregnant: False
Vitals         : Temp=98.0F  Pulse=103bpm SBP=95mmHg  SpO2=93.3%
Symptoms       : chest_pain, diarrhea, none
Risk Score     : 6
Reasoning      : Borderline SpO2; Red-flag symptom(s): chest_pain
AGENT ACTION   -> URGENT_REFERRAL (send to PHC within 24h)

--- Case P006 ---
Age/Gender     : 6 / M | Pregnant: False
Vitals         : Temp=102.3F  Pulse=92bpm SBP=95mmHg  SpO2=97.8%
Symptoms       : headache
Risk Score     : 1
Reasoning      : Mild fever
AGENT ACTION   -> ADVISE_HOME_CARE (routine guidance, no referral)

--- Case P007 ---
Age/Gender     : 59 / F | Pregnant: False
Vitals         : Temp=102.1F  Pulse=101bpm SBP=105mmHg  SpO2=90.6%
Symptoms       : weakness
Risk Score     : 5
Reasoning      : Mild fever; Low SpO2 (<92%)
AGENT ACTION   -> MONITOR 24H (recheck vitals, advise precautions)

--- Case P008 ---
Age/Gender     : 78 / M | Pregnant: False
Vitals         : Temp=102.1F  Pulse=123bpm SBP=178mmHg  SpO2=88.7%
Symptoms       : headache, weakness
Risk Score     : 9
Reasoning      : Mild fever; Low SpO2 (<92%); Abnormal pulse rate; Abnormal blood pressure
AGENT ACTION   -> EMERGENCY ESCALATION (call 108 / teleconsult NOW)

======================================================================
  Session Performance Summary
======================================================================
Total cases screened : 8
  URGENT_REFERRAL          : 2
  EMERGENCY_ESCALATION     : 2
  MONITOR_24H              : 3
  ADVISE_HOME_CARE         : 1
======================================================================
```

Fig 4. Terminal output — 8 simulated patient screenings and session performance summary.

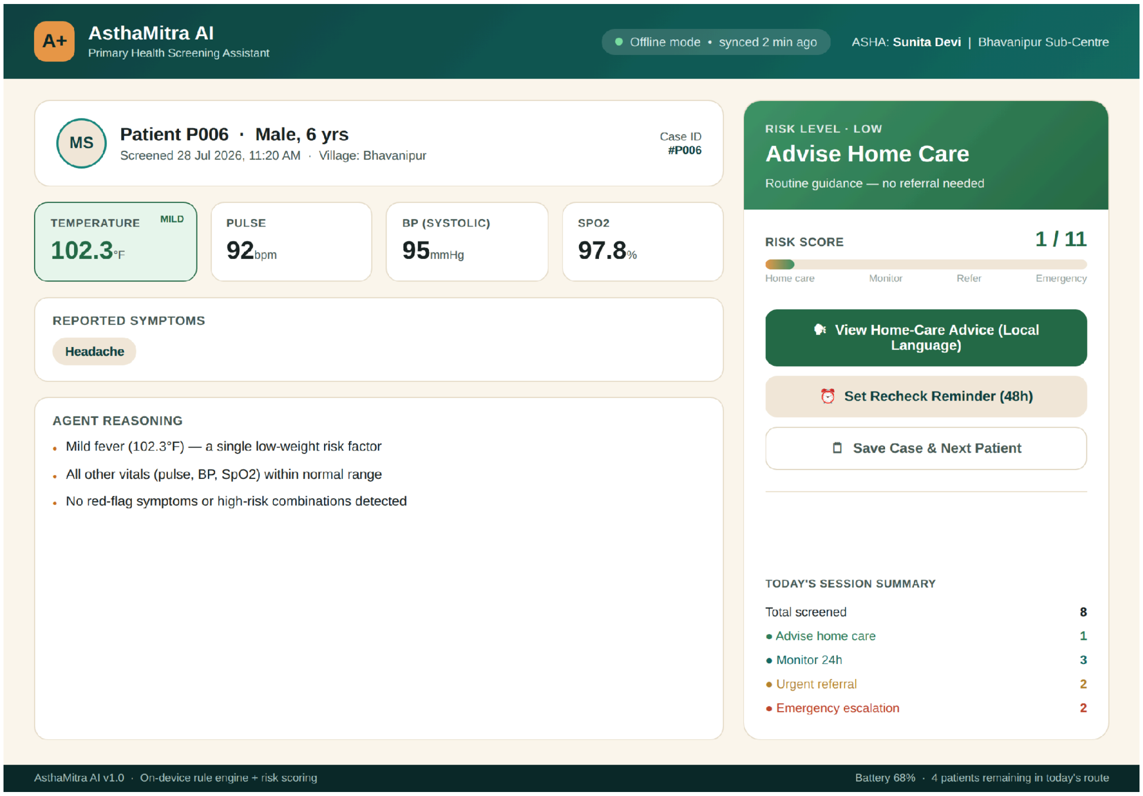
9.3 Demo Output



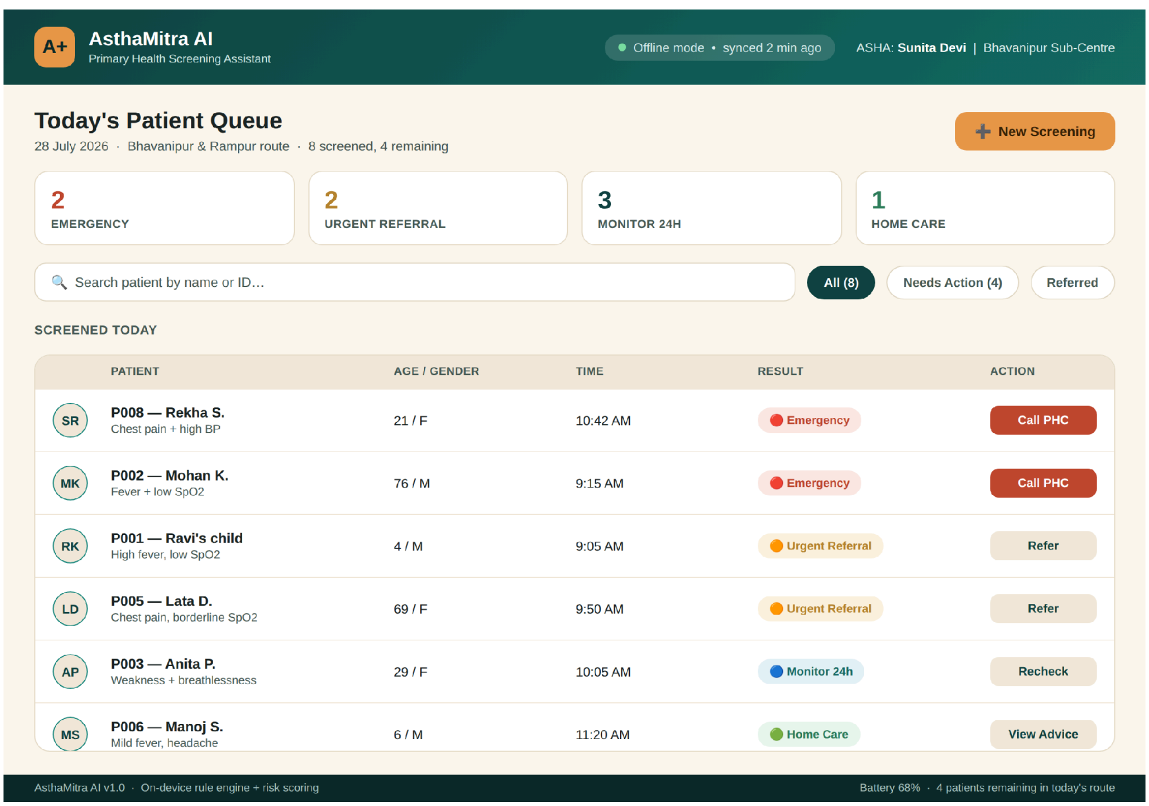
Live demo 1 — Case #P008: Emergency Escalation (risk score 10/11), chest pain + abnormal vitals.



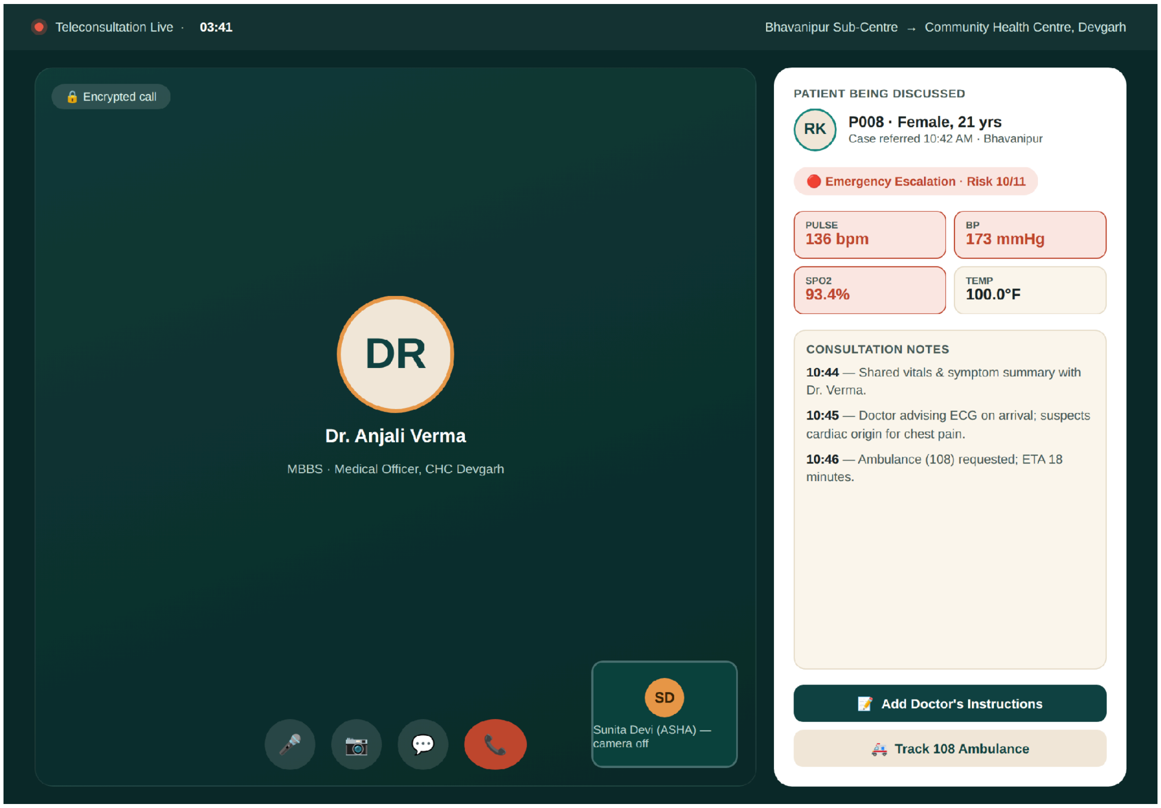
Live demo 2 — Case #P001: Urgent Referral (risk score 7/11), infant with high fever and low SpO2.



Live demo 3 — Case #P006: Advise Home Care (risk score 1/11), mild fever only, all other vitals normal.



Live demo 4 — Today's Patient Queue: dashboard view of all screenings, colour-coded by risk level, with one-tap referral/call actions.



Live demo 5 — Teleconsultation screen: ASHA worker on a live encrypted call with the PHC doctor for Case #P008, with vitals and consultation notes side by side.